

FIRST MEASUREMENT OF CHARGED CURRENT MUON NEUTRINO-INDUCED K^+ PRODUCTION ON ARGON USING THE MICROBOONE DETECTOR

I. DATA/MC COMPARISON FOR SIDEBANDS

This section shows data/MC comparisons for different kinematic variables studied in this analysis. The data used for the comparison is equivalent to a total of 6.88×10^{20} POT (MicroBooNE Run 1-3 datasets).

A near and far sideband region were defined based on a 2D plot of the BDT score vs. daughter track length, as shown in Figure 1. The near and far sideband data/MC comparison shows good agreement (within the statistical uncertainty, see Figure 2 and 3) for kinematic variables studied during the development of this analysis. The agreement with data shows that the background predictions from the MC can be used in the single-bin flux-integrated cross-section measurement without the need to implement a background constraint. The χ^2 value for each distribution in Figures 2 and 3 were calculated using Pearson's χ^2 test, only considering statistical uncertainty.

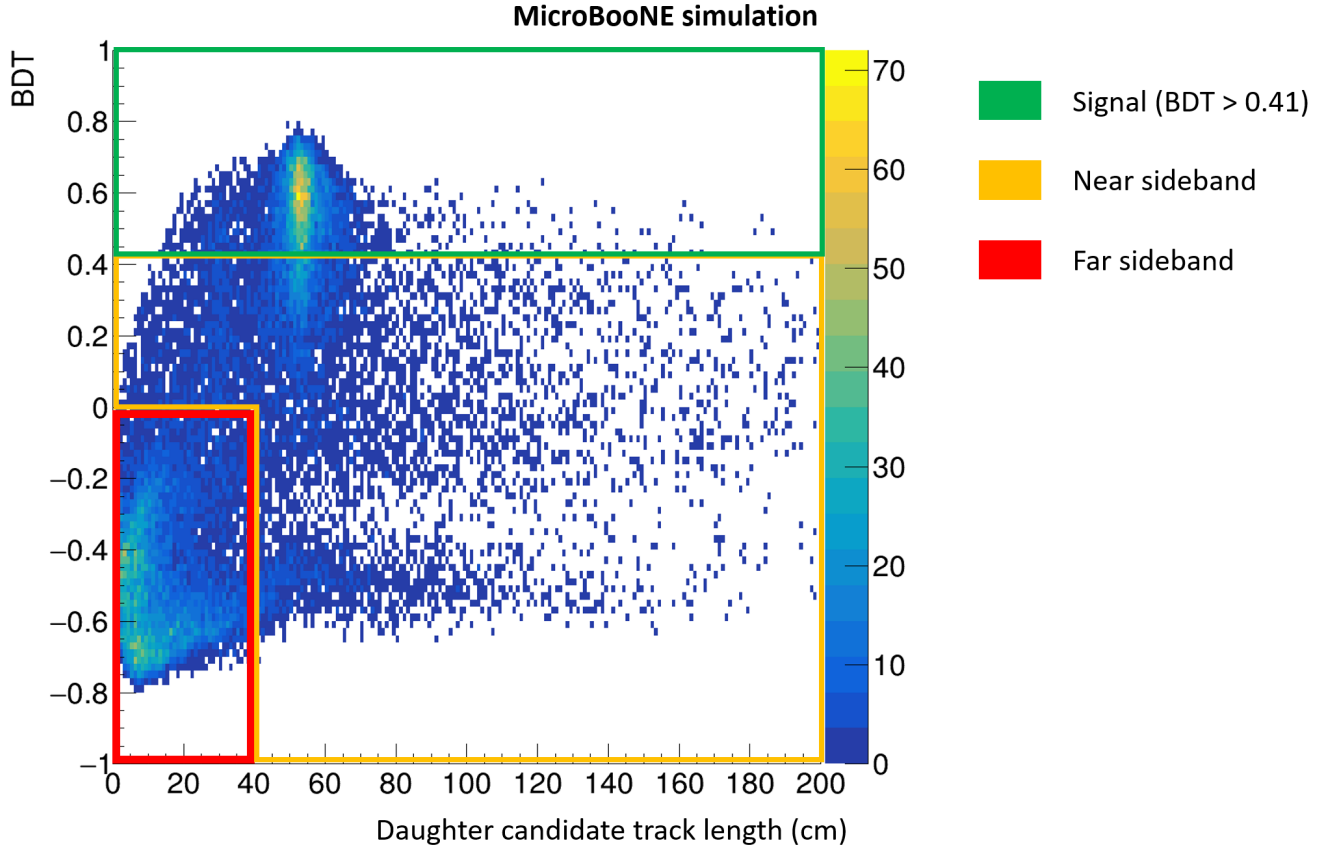


Figure 1: Sideband regions used in this analysis. Signal region (BDT > 0.41, green region), near sideband (yellow region), and far sideband (red region).

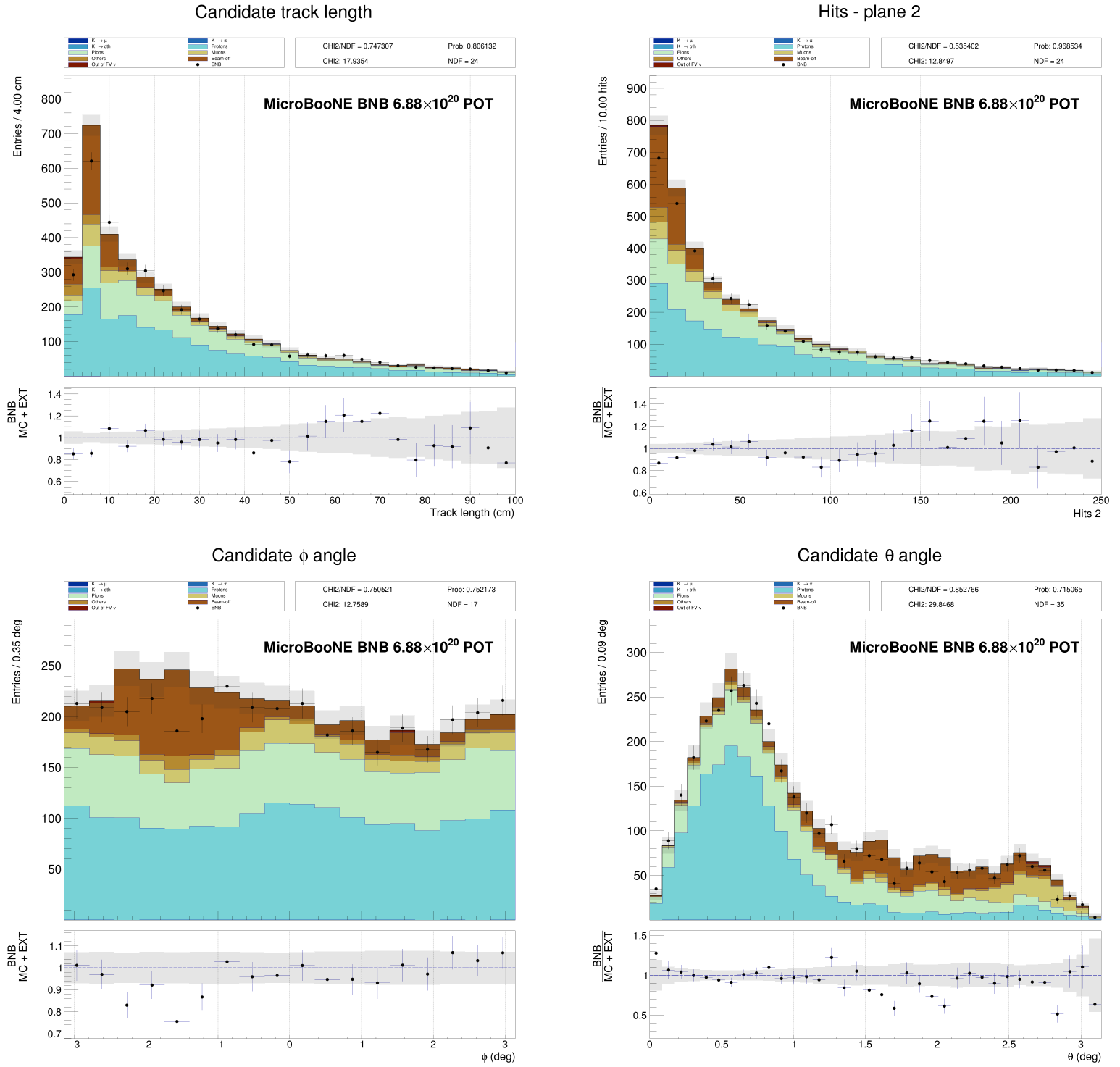


Figure 2: Far sideband data/MC comparisons for track length, ϕ , θ angles, number of hits on collection plane (plane 2).

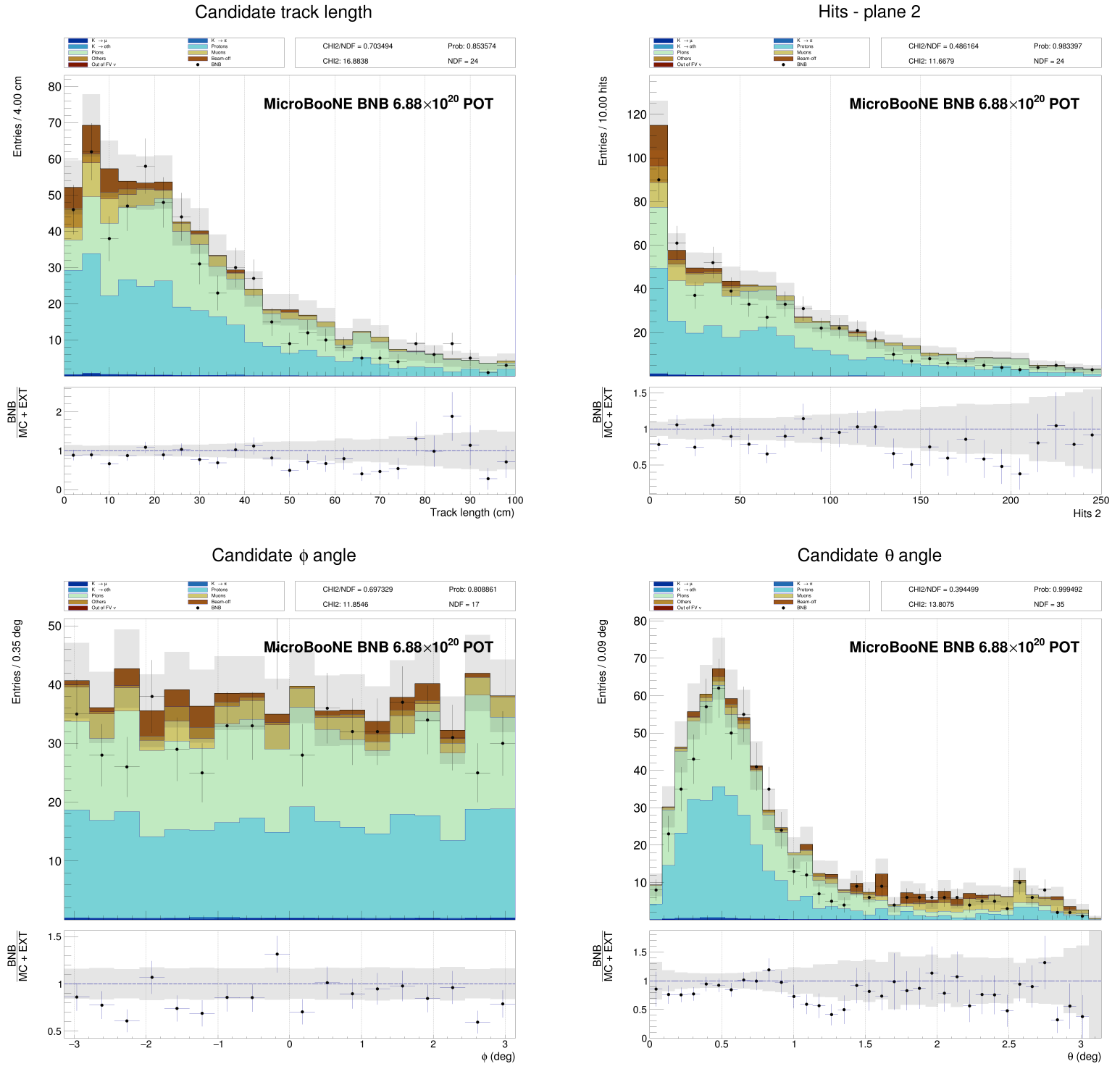


Figure 3: Near sideband data/MC comparisons for track length, ϕ , θ angles, number of hits on collection plane (plane 2).

II. EVENT DISPLAY OF EVENTS SELECTED IN DATA RUN 1-3

This section shows the event displays of the 10 selected kaon candidates in the 6.88×10^{20} POT dataset used for this analysis. For each panel below (Figures 4 and 5), the event displays were generated from the TPC collection plane view. Figure 6 shows the daughter particle track length distribution and the event candidates selected after the BDT cut at 0.41.

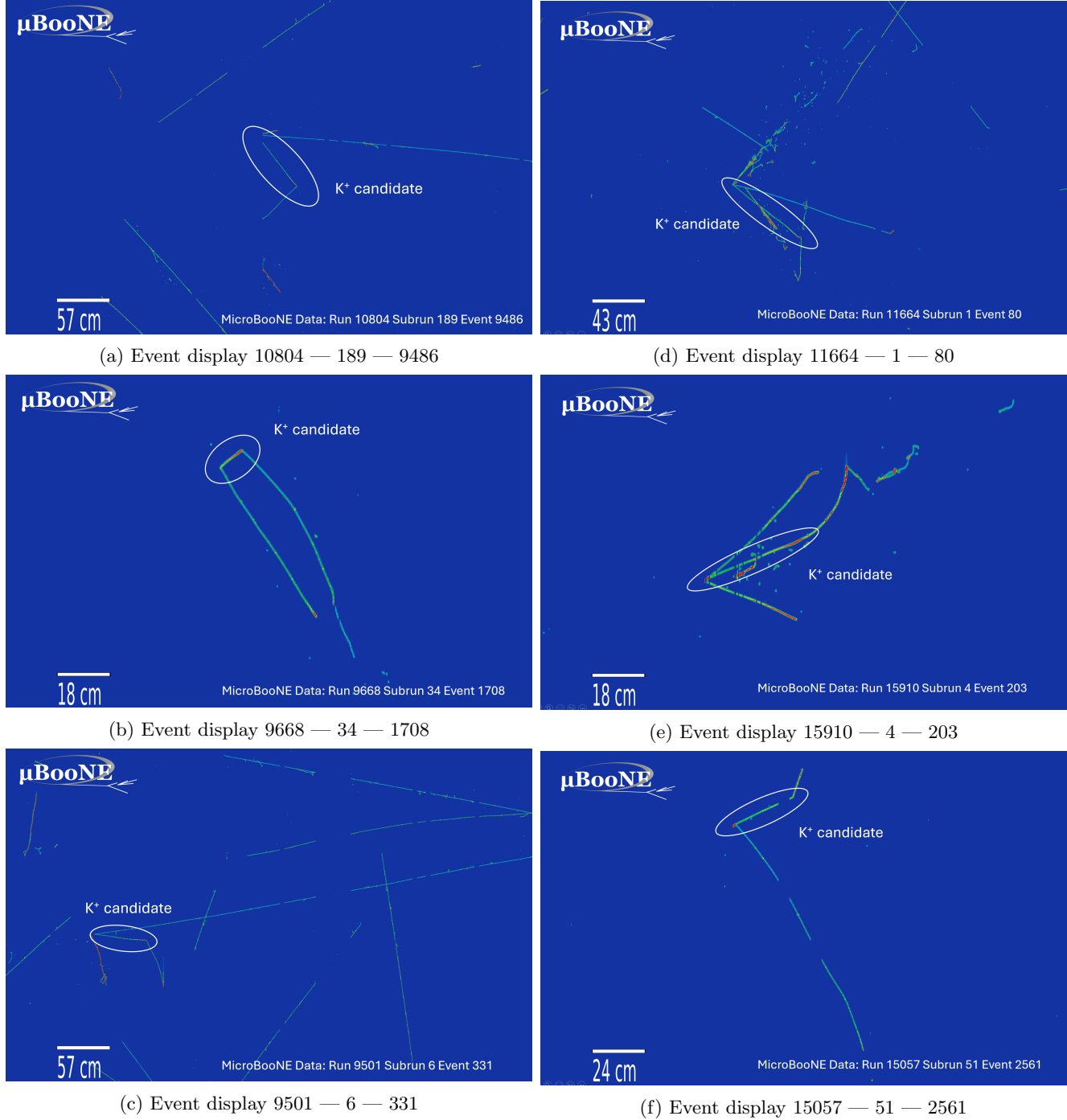


Figure 4: Event display of K^+ candidates.

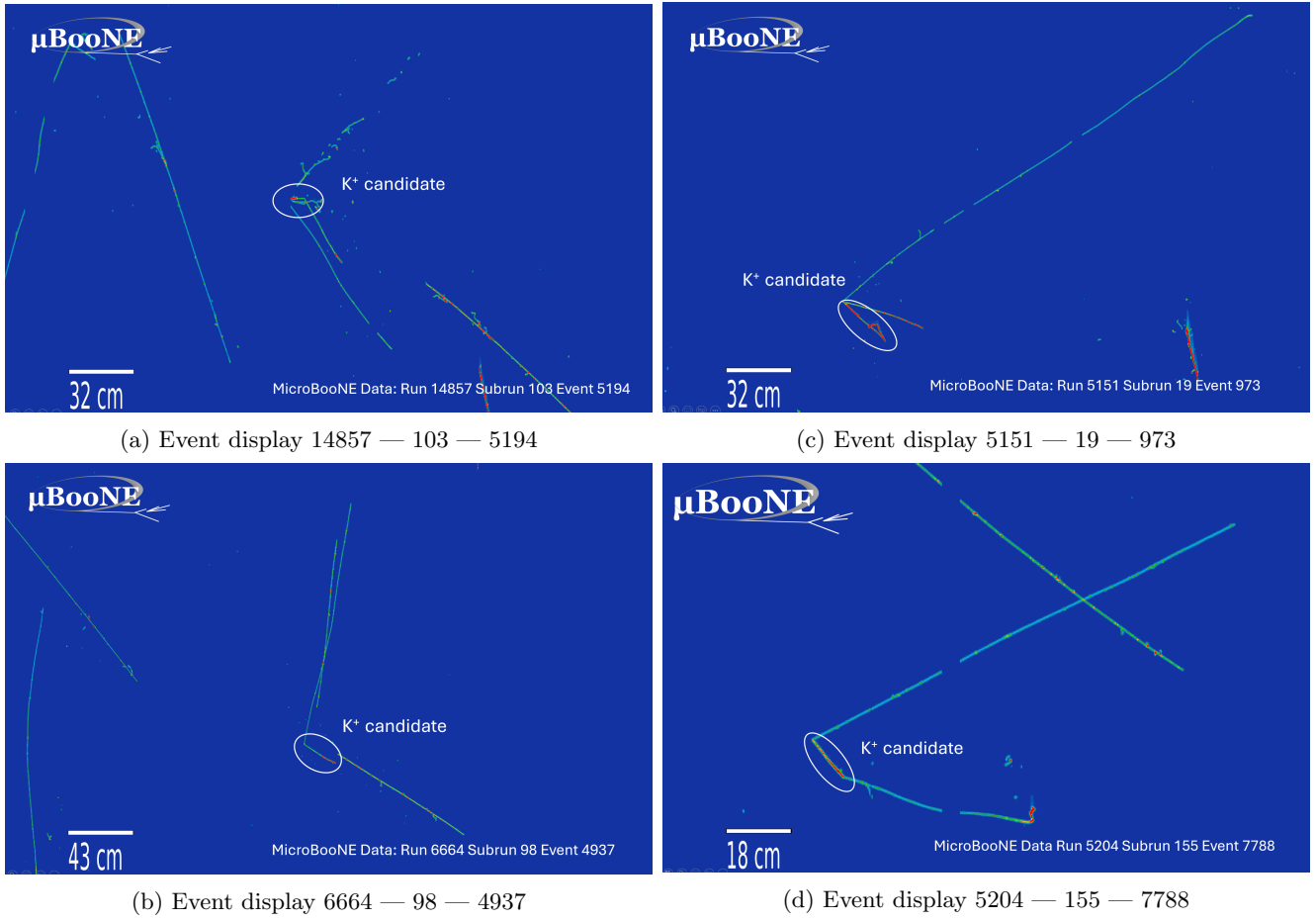


Figure 5: Event display of K^+ candidates.

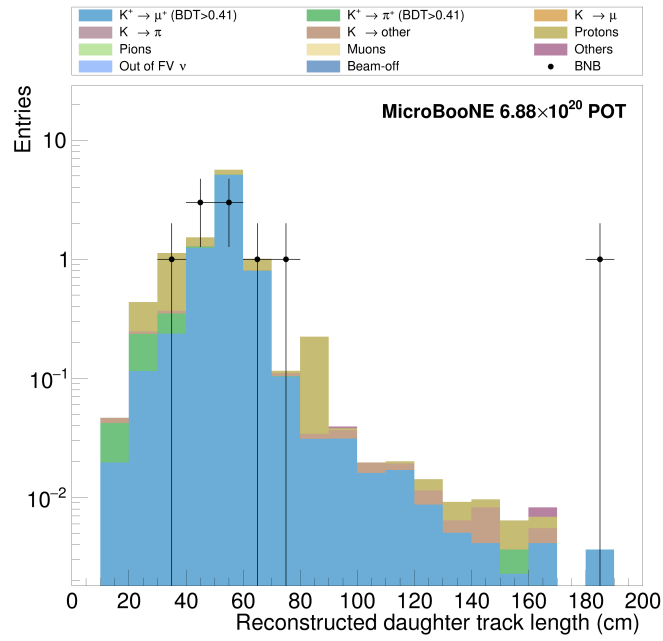


Figure 6: Daughter track length distribution for events selected by the cut $\text{BDT} > 0.41$.

III. EXPECTED BACKGROUND FROM MC IN THE SIGNAL REGION (BDT > 0.41)

This section shows a background event in the signal region. Backgrounds are mainly protons that have attached tracks at their endpoint that mimic a K^+ decay products. Fig. 7 shows an example of an event display of a background event selected in the signal region in the MC BNB sample.

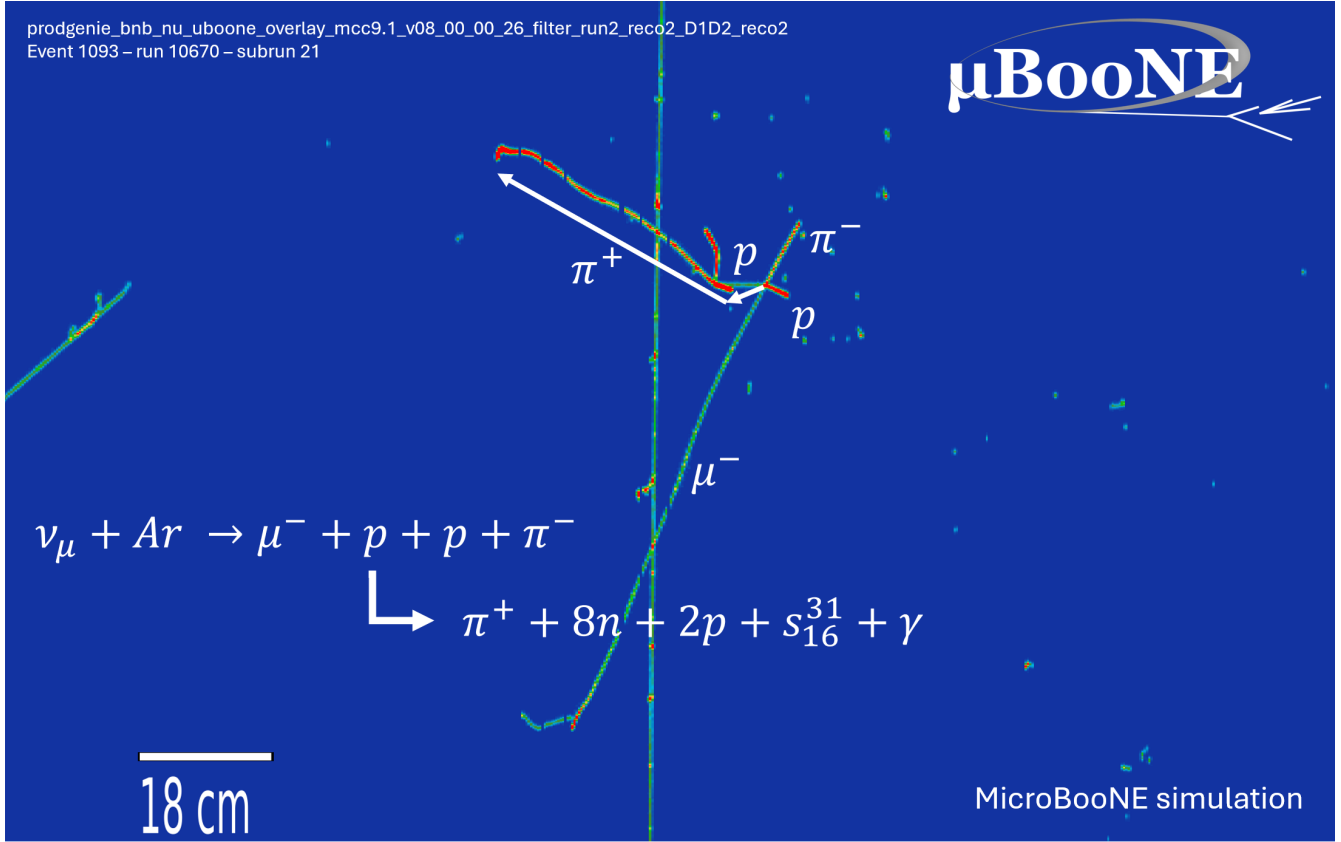


Figure 7: Background event selected in the signal region (BDT > 0.41).

IV. DATA AND GEANT4 CROSS SECTION IN $K^+ - \text{Ca}$ INTERACTIONS

In this section, we compare data with the GEANT4 cross section for $K^+ - \text{Ca}$ interactions. A 40% discrepancy between the available $K^+ - \text{Ca}$ data and the GEANT4 model is observed. We use this to conservatively estimate the $K^+ - \text{Ar}$ reinteraction uncertainty as $\pm 40\%$.

Figure 8 shows the $K^+ - \text{Ca}$ interaction's data [1] (black triangles), the red curve is the GEANT4 $K^+ - \text{Ca}$ cross section as a function of the kinetic energy of the K^+ , and green curves represent $\pm 40\%$ variation on the K^+ inelastic cross-section component to cover the $K^+ - \text{Ca}$ data in low energy region.

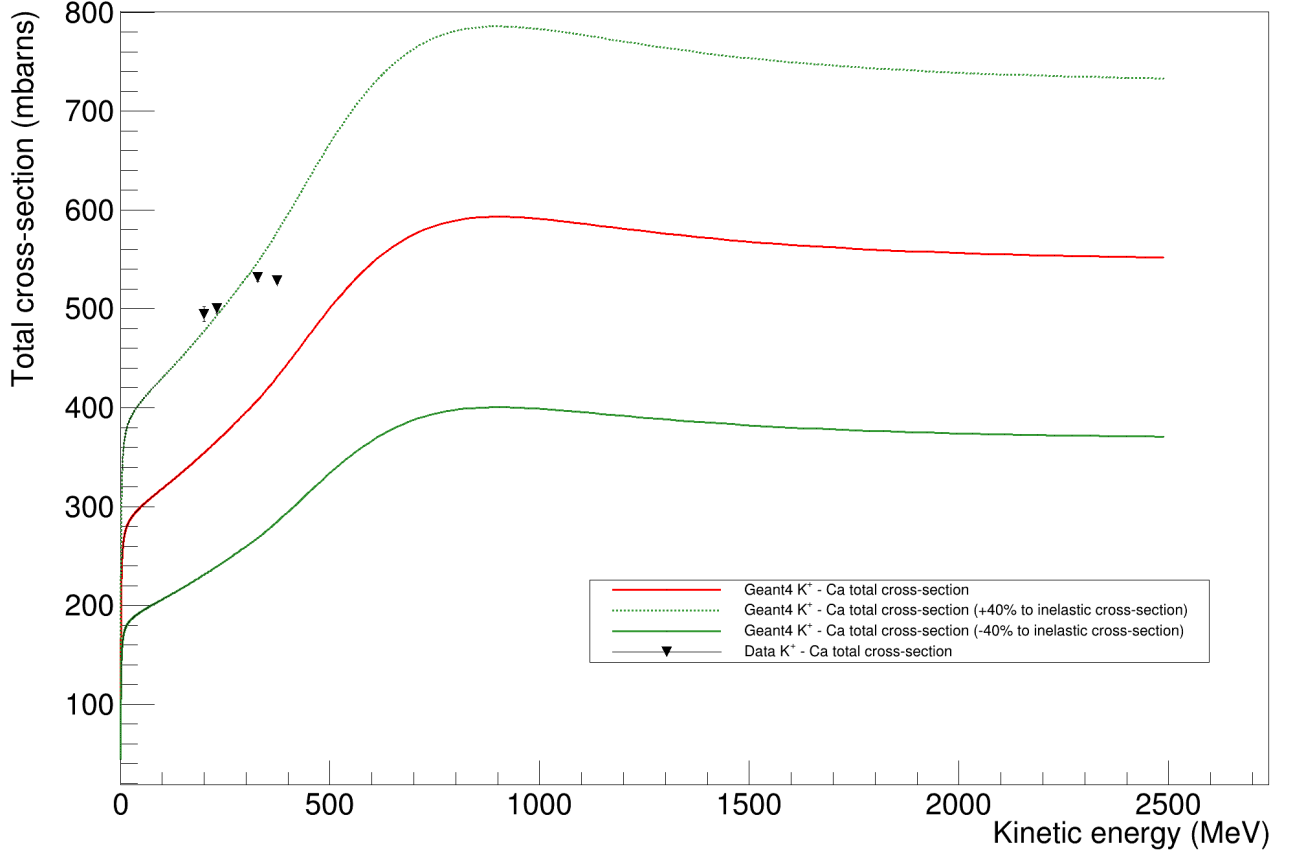


Figure 8: Data and GEANT4 cross section in $K^+ - \text{Ca}$ interactions. Green curves represent a plus (dashed green line) and minus (green line) 40 percent variation of the $K^+ - \text{Ca}$ cross section.

V. PHASE SPACE CONSTRAINT FOR CHARGED-CURRENT K^+ PRODUCTION CROSS SECTION

This section discusses the phase space constraints applied in this analysis. The identification of a region where K^+ can successfully be reconstructed was done by evaluating the efficiency of reconstructed tracks for signal events as a function of K^+ momentum. The efficiency is defined as the ratio of reconstructed K^+ , with attached daughter tracks at the K^+ endpoint and contained within the fiducial volume, to the total number of generated K^+ . It was found that the efficiency drops to zero below 230 MeV and above 2900 MeV, establishing these values as the phase space constraints for this analysis.

Figure 9 shows the final cross section calculated with the MicroBooNE dataset analyzed and reported in Table 1 of the letter. The ν_μ neutrino flux used for the cross-section calculation is provided in the ancillary files, along with the supplemental material and a README file with usage instructions.

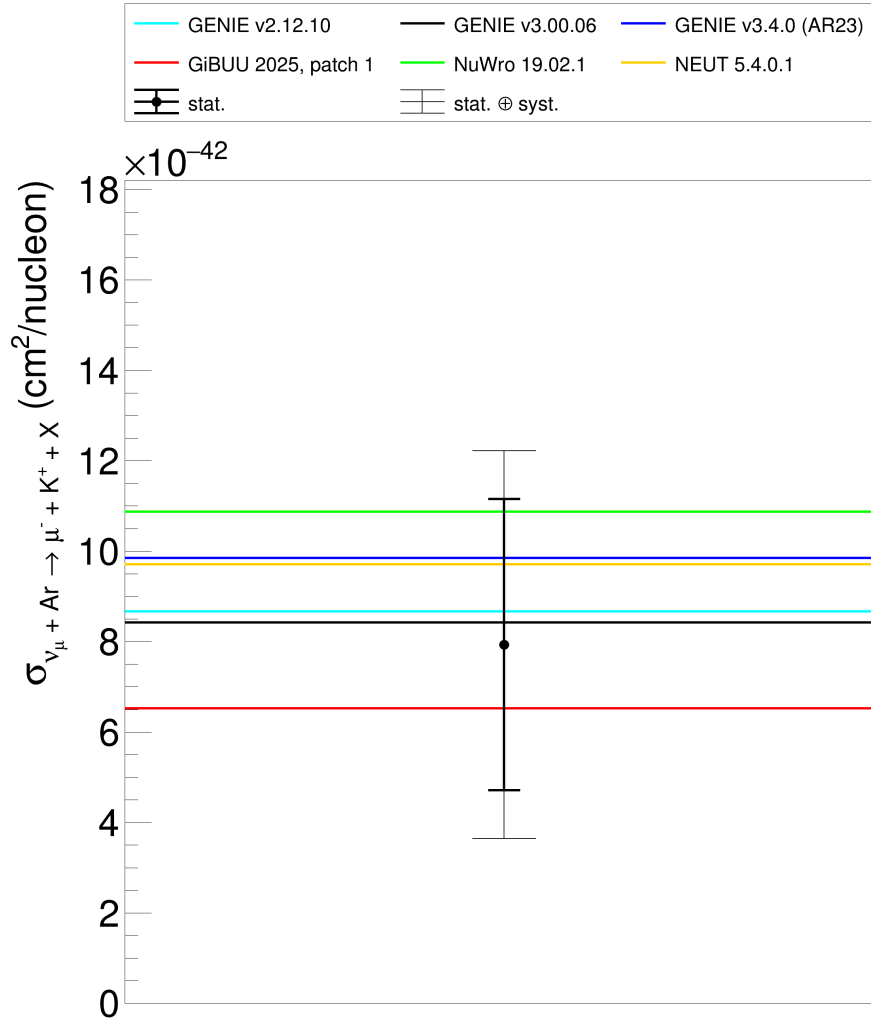


Figure 9: Cross section calculated after applying the phase space constraint. The black dot represents the measurement in data. Color lines are the cross section expected from other neutrino generators, where the phase space constraint was also applied.

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- [1] E. Friedman, A. Gal, R. Weiss, *et al.*, “ K^+ nucleus reaction and total cross sections: New analysis of transmission experiments,” *Phys. Rev. C*, vol. 55, pp. 1304–1311, Mar 1997.